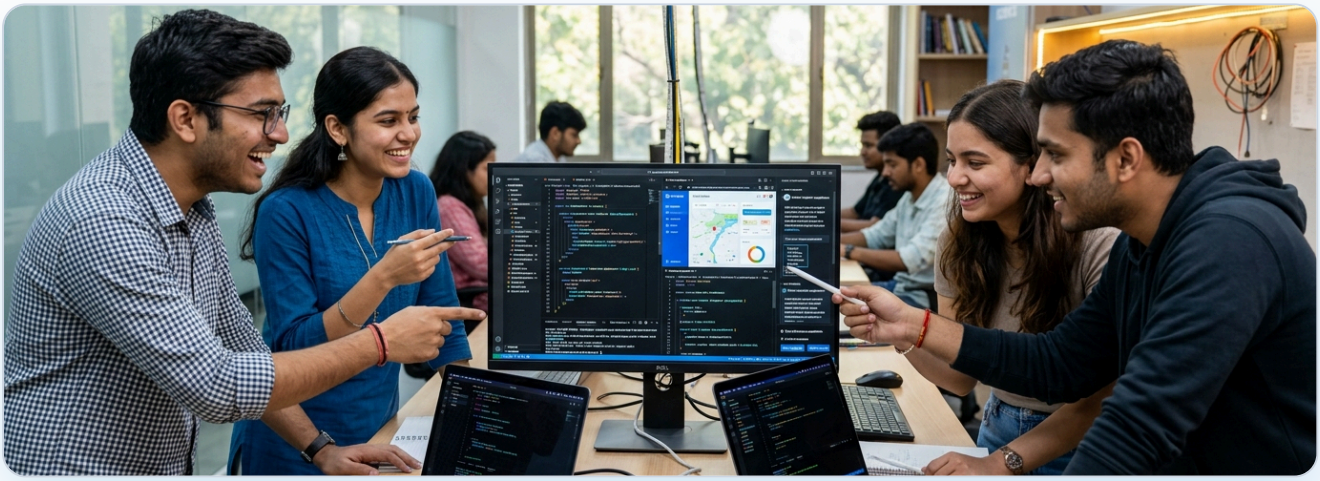
WinNovation

NEP 2020 & TECHNICAL EDUCATION SKILL ELECTIVE

NATIONAL EDUCATIONAL BLUEPRINT • 45-DAY EXPERIENTIAL TRACK

MODERN WEB DEVELOPMENT & FRONTEND ENGINEERING

A comprehensive, industry-aligned STEM & vocational curriculum engineered for Colleges, Polytechnics, Skill Hubs & Advanced Technology Institutes.



DURATION
45 Days
90 Instructional Periods (2 hrs/day)

PEDAGOGICAL SPLIT
70 : 30
70% Hands-on Lab : 30% Theory

TARGET STANDARD
Undergraduate & Diploma
NSQF Level 5/6 Aligned

PRACTICAL OUTPUT
12+ Projects
7 Mini-Projects + 5 Major Capstones

1. Pedagogical Vision & NEP 2020 Strategic Framework

The **Winnovation Modern Web Development Curriculum** transforms undergraduate learners and technology students into productive software engineering creators. Rooted in **Competency-Based Education (CBE)**, students learn semantic architecture, modern responsive design, algorithmic JavaScript logic, asynchronous API handling, and production cloud deployment.

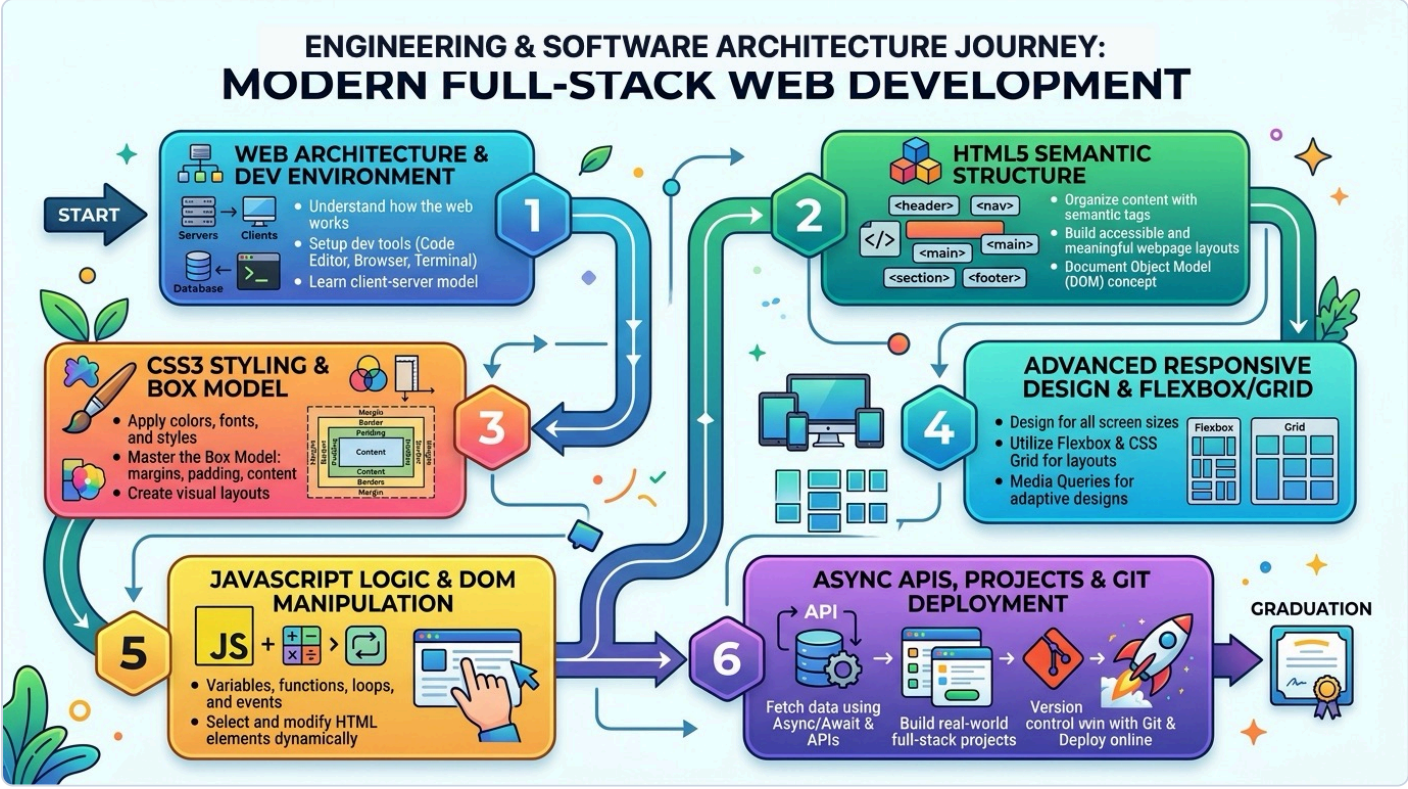
Core Professional Competencies

- **Architectural Thinking:** Client-server separation, DOM tree manipulation, clean modular code.
- **UI/UX & Accessibility:** Mobile-first responsiveness, Flexbox/Grid systems, WCAG compliance.
- **Industry Standard Workflow:** VS Code, Emmet, DevTools debugging, Git version control & Cloud CDNs.

Indian Digital & Enterprise Context

- Prototyping e-commerce storefronts inspired by India's Open Network for Digital Commerce (ONDC).
- Designing accessible government portals, student dashboards, and UPI checkout UI components.
- Strict adherence to web application security and privacy ethics.

2. 6-Stage Experiential Learning Pathway



3. Examination & Assessment Rubric (Institutional Vocational Pattern)

EVALUATION PILLAR	ASSESSMENT INSTRUMENT	WEIGHTAGE	EVALUATION BENCHMARKS & STANDARDS
Continuous Formative Lab	7+ Daily Mini-Project Deliverables	**30 Marks**	Semantic HTML, responsive CSS, event listener hygiene, client-side validation
Mid-Term Milestone Capstones	Portfolio (Day 11) + Landing Page (Day 23)	**20 Marks**	Modern CSS Grid/Flexbox layouts, cross-device responsiveness, UI aesthetic
Comprehensive Major Capstone	Interactive Web App with API & Storage (Days 43-45)	**20 Marks**	Dynamic API fetching, state persistence with LocalStorage, modular ES6+ code, Git repo
Comprehensive Practical Lab Examination	Certified Lab Journal & Live Coding Exam	**15 Marks**	Invigilated hands-on coding, live DOM manipulation task, verified lab journal
Viva Voce & Technical Defense	Oral Examination & Code Defense	**15 Marks**	Client-server mechanics, Event Loop reasoning, CSS specificity, live debugging
TOTAL QUALIFYING AGGREGATE		**100 Marks**	**Minimum Qualifying Benchmark: 40% in Theory & 40% in Practical**

MODULE 1 & 2: WEB ARCHITECTURE & HTML5 FUNDAMENTALS (DAYS 1 – 11)

Environment: VS Code / Chrome DevTools

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 01	**Web Architecture & Protocols** • How the Web Works, DNS, IP Addresses • Client-Server Model, HTTP/HTTPS Request-Response • Static vs. Dynamic Websites	Trace packet hops using network utilities; inspect HTTP status codes (200, 404, 500) and headers.	Mini Project **Web Architecture Primer** Annotated client-server architectural diagram & request flow log.
Day 02	**Web Browsers & Developer Tools** • Rendering engines (Blink, Gecko, WebKit) • Chrome DevTools: Elements, Console, Network tabs • Simulating mobile viewports and network throttling	Audit an live website using DevTools: edit DOM live, inspect computed styles, monitor asset download waterfall.	Mini Project **Browser DevTools Audit Sheet** Diagnostic performance and asset weight audit report.
Day 03	**Professional Development Environment** • VS Code setup, integrated terminal, Prettier • Live Server extension, file tree organization • Emmet abbreviations syntax (`!` , `ul>li*5`)	Initialize a structured multi-page workspace folder; practice 20 essential Emmet expansion shortcuts.	Mini Project **Configured Developer Workspace** Verified project environment + Emmet productivity reference sheet.
Day 04	**HTML5 Skeleton & Metadata** • HTML syntax rules, tags, elements, attributes • `<!DOCTYPE html>` , `<html>` , `<head>` , `<body>` • Meta tags: charset, viewport, SEO title/description	Author a clean HTML5 boilerplate from scratch with valid meta tags and social share open-graph tags.	Mini Project **Standards Boilerplate Template** W3C-validated HTML5 starter document template.
Day 05	**Text Hierarchy & Typographic Formatting** • Headings (`<h1>` to `<h6>`), Paragraphs (`<p>`) • Formatting: `<strong>` , `<em>` , `<mark>` , `<small>` • `<blockquote>` , `<hr>` , ` ` , character entities	Structure an informational news editorial about India's space missions (ISRO) with typographic hierarchy.	Mini Project **Editorial News Article Page** Typographically structured editorial article.
Day 06	**Lists & Content Classification** • Unordered lists (`<ul>`), Ordered lists (`<ol>`) • Description lists (`<dl>` , `<dt>` , `<dd>`) • Nested lists for multi-level navigation trees	Construct a course syllabus directory and a technical computer science glossary using nested and description lists.	Mini Project **Syllabus Directory & Glossary** Multi-level list structure and technical glossary webpage.
Day 07	**Hyperlinks & Resource Navigation** • Anchor tag (`<a>`), `href` , Relative vs. Absolute URLs • `target="_blank"` , security with `rel="noopener"` • Page fragment bookmarks (`#section-id`), mailto, tel	Build a 4-page institutional portal with working inter-page navigation menus and smooth jump-to-top bookmarks.	Mini Project **Multi-Page Institutional Portal** Interconnected 4-page navigable website structure.
Day 08	**Multimedia: Images, Audio & Video** • `<img>` , accessibility `alt` , `<figure>` , `<figcaption>` • Responsive art direction with `<picture>` and `srcset` • Native HTML5 `<audio>` and `<video>` controls	Build a multimedia media gallery featuring optimized image formats, audio previewer, and video players.	Mini Project **Multimedia Showcase Gallery** Accessible media gallery featuring audio, video, and pictures.
Day 09	**Tabular Data & Table Architecture** • `<table>` , `<tr>` , `<th>` , `<td>` , `<caption>` • Structural groups: `<thead>` , `<tbody>` , `<tfoot>` • Complex merging: `colspan` , `rowspan` , scope	Construct a comprehensive University Academic Examination Timetable and Grade Matrix with merged subject cells.	Mini Project **Academic Timetable & Grade Matrix** W3C-compliant complex HTML table with cell spans.
Day 10	**Interactive Forms & Input Elements** • `<form>` , `action` , `method` , `<label>` associations • Input types: text, email, password, date, radio, check • `<select>` , `<textarea>` , `required` , `pattern` regex	Build an Indian College Admission Application Form with client-side field validation constraints.	Mini Project **Student Admission Form** Validated user registration and admission form.

MODULE 2 MILESTONE CAPSTONE • DAY 11

Personal Portfolio & Digital Resume Website (Semantic HTML5)

Semantic Architecture: Build a clean, multi-section personal portfolio website utilizing strict semantic elements: `<header>` , `<nav>` , `<main>` , `<section>` , `<article>` , `<aside>` , and `<footer>` .

Sections: About Me, Educational Qualifications (Table), Skill Sets (Lists), Multimedia Project Showcase, and Contact Form.
Deliverable: W3C-validated, semantic HTML5 digital resume and portfolio (`index.html`).

MODULE 3 & 4: CSS3 STYLING, BOX MODEL & 1D/2D LAYOUTS (DAYS 12 – 23)

Modern CSS3 / Flexbox / CSS Grid

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 12	**Introduction to CSS & Syntax**: Rule anatomy, Inline vs. Internal vs. External stylesheets (`<link>`), The Cascade, Inheritance, Reset/Normalize CSS.	Link an external stylesheet to unstyled HTML; observe cascading precedence and inheritance rules.	Mini Project **External Styled Document**: Cleanly styled documentation page.
Day 13	**CSS Selectors & Specificity Hierarchy**: Element, Class, ID, Grouping, Combinators (descendant, child), Pseudo-classes (`:hover` , `:focus` , `:active` , `:nth-child`).	Style an interactive button deck and navigation list using specificity rules and pseudo-classes.	Mini Project **Interactive Button & Nav Set**: Styled navigation components with hover states.
Day 14	**Colors, Backgrounds & Borders**: Named, HEX, RGB, RGBA, HSL, Background image, cover, contain, Borders, `border-radius` , Box shadows (`box-shadow`).	Create elevated product cards with rounded corners, subtle shadows, and gradient hero banners.	Mini Project **Modern Product Card UI**: Elevated card components with shadows.
Day 15	**Web Typography & Google Fonts**: `font-family` , Google Fonts integration (`@import` , `<link>`), `line-height` , `letter-spacing` , `text-transform` .	Pair primary and accent Google Fonts on an editorial layout to achieve balanced reading typography.	Mini Project **Editorial Typography Sheet**: Harmonious typographic article layout.
Day 16	**The CSS Box Model Masterclass**: Content, Padding, Border, Margin; Margin collapsing; `box-sizing: content-box` vs. `border-box` .	Construct nested layout boxes; audit spacing using Chrome DevTools Box Model visual inspector.	Mini Project **Box Model Diagnostic Page**: Visual spacing calibration test page.
Day 17	**Dimensions & Relative Sizing Units**: Absolute (`px`) vs. Relative (`%` , `em` , `rem` , `vw` , `vh`); `max-width` , `min-height` for flexible containers.	Build a fluid content container that dynamically scales within minimum and maximum boundaries.	Mini Project **Fluid Container System**: Scalable layout with relative units.
Day 18	**Display Properties & Document Flow**: Normal flow, `display: block` , `inline` , `inline-block` , `display: none` vs. `visibility: hidden` .	Convert vertical navigation lists into horizontal menu bars using `inline-block` .	Mini Project **Horizontal Menu Bar**: Clean inline-block navigation bar.
Day 19	**CSS Positioning & Stacking Context**: `position: static` , `relative` , `absolute` , `fixed` , `sticky` ; Offsets; `z-index` stacking order.	Build a sticky navigation bar, a floating action button (FAB), and an absolute badge on an image card.	Mini Project **Sticky Header & Floating UI**: Multi-layer positioned web interface.
Day 20	**Flexbox 1D Layout (Container Properties)**: Main vs. Cross axis, `display: flex` , `flex-direction` , `justify-content` , `align-items` , `flex-wrap` , `gap` .	Construct an aligned multi-item header and a wrapping multi-card product row using Flexbox.	Mini Project **Flexbox Header & Product Row**: Balanced 1D flexbox layout.
Day 21	**Flexbox Item Properties & Alignment**: `flex-grow` , `flex-shrink` , `flex-basis` , shorthand `flex` , `align-self` , `order` .	Build a split-screen hero banner and a flexible messaging chat UI with dynamic message widths.	Mini Project **Flexbox Messaging UI**: Responsive chat interface with auto-growing bubbles.
Day 22	**CSS Grid 2D Layout Fundamentals**: Grid container, Columns, Rows, Fractional units (`fr`), `gap` , `repeat()` , `minmax()` .	Create an auto-fitting multi-column photo and project gallery grid that reflows automatically.	Mini Project **Auto-Fit Responsive Grid**: Modern 2D CSS grid gallery.

MODULE 4 MILESTONE CAPSTONE • DAY 23

Responsive Product Landing Page (Flexbox & CSS Grid)

Layout Architecture: Holy Grail layout combining Flexbox (Header, Nav, Footer) and CSS Grid (`grid-template-areas` for Features, Pricing Table, Testimonials).
Features: Sticky Navbar, Hero with Call-to-Action, 3-tier Pricing Grid, Testimonial cards, and responsive Footer.
Deliverable: Fully styled, responsive promotional landing page (`landing-page.html` + `styles.css`).

MODULE 4 (CONT.): RESPONSIVE DESIGN, MEDIA QUERIES & UI MOTION (DAYS 24 – 27)

CSS Variables / Media Queries / Animations

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 24	**Media Queries & Mobile-First Design**: Viewport meta, Mobile-first strategy, Breakpoints (Mobile < 768px, Tablet 768–1024px, Desktop > 1024px), Responsive Navbar.	Refactor the landing page to seamlessly adapt across mobile, tablet, and desktop viewports.	Mini Project **Multi-Device Adaptive Portal**: Responsive website across 3 breakpoints.
Day 25	**CSS Variables & Dark Theme Architecture**: Defining variables in `:root`, `var()`, Scope, Theme toggle architecture using data attributes.	Build a theme switcher toggling between Light and Dark mode using CSS custom properties.	Mini Project **Themeable Web Interface**: Dark/Light mode theme system.
Day 26	**Visual Styling & Micro-Transitions**: Multi-layer shadows, Glassmorphism with `backdrop-filter`, CSS Transitions (`ease-in-out`).	Build frosted-glass cards with smooth hover transitions and glowing accent borders.	Mini Project **Glassmorphism Component Kit**: Polished UI cards with micro-transitions.
Day 27	**2D/3D Transforms & Keyframe Animations**: `transform` (scale, rotate, translate), `@keyframes` syntax, animation duration, iteration.	Create pulsing CTA buttons, interactive card flips, and a spinning loading spinner.	Mini Project **UI Animation Library**: CSS keyframe micro-interaction suite.

MODULE 5: JAVASCRIPT PROGRAMMING FUNDAMENTALS (DAYS 28 – 35)

Core Language Logic & Algorithmic Foundations

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 28	**JavaScript Foundations & Environment**: V8 engine, Script inclusion (`defer`, `async`), Console methods, Variables: `var` vs. `let` vs. `const`.	Verify variable scoping and re-declaration rules; inspect output in browser console.	Mini Project **Scoping Lab Notebook**: Script analyzing variable lifetimes and TDZ.
Day 29	**Data Types & Type Conversion**: Primitives vs. Reference types, Type coercion, Explicit conversion (`Number()`, `String()`), `typeof`, `==` vs. `===`.	Build a type evaluation script testing truthy/falsy coercion edge cases.	Mini Project **Type Diagnostic Script**: Verified type evaluation module.
Day 30	**Operators & Expressions**: Arithmetic, Assignment, Comparison, Logical (`&&`, `||`, `!`), Short-circuit evaluation, Nullish coalescing (`??`).	Implement a commercial GST and tiered discount pricing calculator.	Mini Project **Commercial Pricing Engine**: Multi-rule transaction pricing calculator.
Day 31	**Control Flow & Branching**: `if-else` chains, Nested conditions, `switch-case` with break/default, Ternary operator.	Build an automated academic grading matrix and dynamic transit ticketing fare calculator.	Mini Project **Decision Logic Module**: Multi-branch decision engine.
Day 32	**Loops & Iterations**: `for`, `while`, `do-while` loops, Loop control (`break`, `continue`), Iterating collections.	Generate multiplication matrices and search prime number sequences algorithmically.	Mini Project **Mathematical Computation Engine**: Loop-based algorithmic script.
Day 33	**Functions & Modularity**: Declarations vs. Expressions, Parameters, Default arguments, `return` statement, Pure functions.	Author a modular financial utility library (`computeEMI`, `computeCompoundInterest`).	Mini Project **Financial Utility Library**: Reusable mathematical functions.
Day 34	**Scope, Hoisting & Execution Context**: Global, Function, Block scope; Scope chain, Hoisting mechanics, Call Stack execution.	Debug scoping shadow bugs and step through execution using the Chrome DevTools debugger.	Mini Project **Call Stack Debug Report**: Documented execution trace analysis.

MODULE 5 MILESTONE CAPSTONE • DAY 35

Interactive Calculator Application (Arrays, Objects & Logic)

Core Logic: Object model managing state (`operand1`, `operator`, `operand2`, `displayValue`). Support for basic arithmetic, decimal point handling, clear (`AC`), backspace, and percentage calculations.

Array Iteration: Render calculator keypads dynamically from array configurations using `map()`.

Deliverable: Fully operational interactive calculator application (`calculator.html` + `script.js`).

MODULE 6: DOM MANIPULATION, EVENTS, ES6+, ASYNC & STORAGE (DAYS 36 – 42) DOM API / Fetch API / Web Storage

DAY	CORE THEORETICAL TOPICS	GUIDED LABORATORY PRACTICAL	MINI PROJECT & DELIVERABLE
Day 36	**DOM Tree Architecture & Manipulation** • DOM Tree: Nodes, Elements, Attributes • Selectors: `getElementById`, `querySelector`, `querySelectorAll` • Modifying content: `textContent`, `innerHTML`, `classList` API	Dynamically generate a multi-item product showcase card list from a raw JavaScript array of data objects.	Mini Project **Dynamic DOM Catalog Renderer** JavaScript script generating dynamic product cards.
Day 37	**Event Handling & Event Delegation** • `addEventListener`, click, input, submit, keydown • Event Object: `e.target`, `e.preventDefault()` • Event propagation: Bubbling, Capturing, Delegation	Build an interactive shopping list where items can be dynamically added, struck out, and deleted via event delegation.	Mini Project **Dynamic Shopping Cart UI** Interactive item manager using delegated events.
Day 38	**Client-Side Form Validation & UI Feedback** • Form submit event intercept, validation logic • Regular expressions (Regex) for Email & Phone • Real-time visual feedback: error borders, helper text	Construct a user registration form with live password strength meter, email regex checking, and submit locking.	Mini Project **Client-Side Validation Form** Bulletproof form validation with live UI feedback.
Day 39	**Modern ES6+ Features Masterclass** • Arrow functions (`() ⇒ {}`), Template literals (`` `...` ``) • Destructuring assignment (Objects & Arrays) • Spread operator (`...`), Rest parameters, Default values	Refactor legacy procedural ES5 code into concise, modern ES6+ syntax using destructuring and arrow functions.	Mini Project **ES6+ Modernized Code Module** Clean, modern JavaScript refactored codebase.
Day 40	**Asynchronous JS: Promises & Async/Await** • Event loop, Call stack, Web APIs, Task queue • `Promise` lifecycle: Pending, Fulfilled, Rejected • Modern `async / await` syntax with `try...catch`	Simulate asynchronous multi-step food order processing with promises, timers, and error recovery.	Mini Project **Async Order Simulator** Simulation script managing delayed execution.
Day 41	**Fetch API, JSON & External REST Services** • JSON format: `JSON.stringify()`, `JSON.parse()` • REST API architecture, HTTP methods (GET, POST) • `fetch()` requests, Response headers, Network errors	Connect to public OpenWeatherMap API: fetch real-time temperature, weather icons, and city forecasts.	Mini Project 7 **Live Weather App using API** Dynamic weather app rendering real-time API data.

MODULE 6 MILESTONE CAPSTONE • DAY 42

Browser Storage & State Persistence: To-Do List & Quiz App

Web Storage Integration: Master `localStorage` vs. `sessionStorage` to persist complex JavaScript object arrays across browser sessions.

• **Mini Project 5 (Persistent To-Do App):** Add tasks, mark complete, filter (All, Active, Completed), persist state to LocalStorage.

• **Mini Project 6 (Interactive Quiz App):** Timed multiple-choice quiz, score calculation, instant feedback, high-score leaderboard stored in LocalStorage.

Deliverables: 2 fully functional, persistent web applications (`todo-app.html` and `quiz-app.html`).

MODULE 7: MAJOR CAPSTONE, VERSION CONTROL & CLOUD DEPLOYMENT (DAYS 43 – 45) Full-Stack Frontend Showcase

STAGE / DAY	INSTRUCTIONAL SCOPE & CORE ACTIVITY	GUIDED EXECUTION TASKS	CAPSTONE DELIVERABLE
Day 43 **Major Project**	Architecture of enterprise web applications: Choose 1 of 5 tracks (E-Commerce Storefront, Enterprise Resource Dashboard, Job Portal, Food Ordering, Travel Experience Portal).	Implement multi-page routing, state persistence (LocalStorage), dynamic product search/filter, and modal dialogs.	**Production Major Web Application** Feature-complete responsive web application with interactive logic and state management.
Day 44 **Git & GitHub**	Git version control: Repositories, Staging, Commits, Branches, GitHub remote connection, Push/Pull, Markdown README authoring.	Initialize local Git repository; write structured `README.md` with architecture diagrams, screenshots, and live demo links.	**Structured GitHub Repository** Public GitHub repository with clean commit history and comprehensive documentation.
Day 45 **Deployment**	Modern Jamstack cloud deployment on Vercel, Netlify, and GitHub Pages. CI/CD automated build pipelines, custom domains.	Deploy production web application to Netlify/Vercel; verify SSL certificate; present final portfolio to evaluation committee.	**Live Production URL & Technical Defense** Live hosted application link + certified project dossier + Viva Voce defense.

Major Project Track Selection (Choose One for Days 43–45):

- **Track 1 — E-Commerce Storefront:** Category filtering, live search, dynamic cart drawer, LocalStorage checkout.
- **Track 2 — Enterprise Performance & Resource Dashboard:** Department registry, attendance ledger, dynamic performance index calculation, interactive filter tables.
- **Track 3 — Job Search Portal:** Searchable listings, salary filter, bookmarked favorites, application modal.
- **Track 4 — Food Ordering App UI:** Restaurant menus, cart calculation, delivery address validator.
- **Track 5 — Travel & Tourism Experience Portal:** Destination guide, interactive trip planner, itinerary builder.

4. Academic Lab Specifications & Web Ethics Guidelines

Recommended Enterprise IT Lab Infrastructure:

- Hardware: Dual-Core Intel i3/i5 or AMD Ryzen 3/5, 8GB RAM, 100GB Free Disk
- Operating System: Windows 10/11 64-bit or Ubuntu 22.04 LTS or macOS
- Code Editor: Visual Studio Code with Live Server, Prettier, Auto Rename Tag
- Browsers: Google Chrome & Mozilla Firefox with Developer Tools
- Version Control: Git 2.x Command Line & GitHub Desktop Client

Web Ethics & Cyber Security Guidelines:

- All production web applications must sanitize form inputs to prevent XSS attacks.
- Strict adherence to accessibility guidelines (Semantic tags, alt text).
- Enforce secure HTTPS protocols and avoid storing plain-text passwords.
- Career & Technical Pathways: Full-Stack Engineer, Frontend Architect, Cloud Systems Developer.